

Project 1

Factors Affecting Log Wage

Regression Analysis

Member 1: Le Ngoc Khoa - SE203900
Member 2: Tran Phu Tan - SE203895
Course: MAI391 - Mathematics for Machine Learning
Instructor: Tran Trong Huynh
Date: July 21, 2026
Repository: [GitHub Link](#)



FPT UNIVERSITY

All analyses performed in Python (pandas, statsmodels, scikit-learn).

Contents

1	Introduction	3
2	Data Description	3
2.1	Dataset Overview	3
2.2	Missing Values	3
3	Exploratory Data Analysis (EDA)	4
3.1	Descriptive Statistics	4
3.2	Distribution of Log Wage	4
3.3	Scatter Plots	5
3.4	Correlation Analysis	5
4	ANOVA Analysis	6
5	OLS Regression Analysis	7
5.1	Model Specifications	7
5.2	Model Fit Comparison	8
5.3	Incremental F-Tests	8
5.4	Coefficient Estimates and Interpretation	8
6	Machine Learning Models	9
6.1	Data Splitting and Preprocessing	9
6.2	Models	9
6.3	Performance Comparison	10
6.4	Feature Selection by Lasso	10
7	Model Diagnostics	11
7.1	Normality of Residuals	11
7.2	Heteroscedasticity	11
7.3	Multicollinearity (VIF)	12
7.4	Autocorrelation (Durbin-Watson)	12
8	Conclusions and Recommendations	12
8.1	Summary of Findings	12
8.2	Limitations	12
8.3	Recommendations	13
	References	13
A	Python Code Summary	13

List of Figures

1	Distribution of Log Wage: Histogram, Q-Q Plot, and Boxplot	5
2	Scatter Plots: Continuous Predictors vs Log Wage	5
3	Pearson Correlation Matrix Heatmap	6
4	Wage Distribution by Group — One-Way ANOVA Boxplots	7
5	ML Model Performance: RMSE, R^2 , Cross-Validation, and Predicted vs Actual	10
6	Residual Diagnostic Plots for Model 3	11

List of Tables

1	Variable Descriptions	3
2	Missing Values Before Cleaning	4
3	Descriptive Statistics ($n = 1,603$)	4
4	One-Way ANOVA Results ($\alpha = 0.05$)	7
5	OLS Model Comparison	8
6	OLS Regression Results — Model 3 (Full Model)	9
7	ML Model Performance Metrics	10

1. Introduction

This report investigates the determinants of workers' wages using regression analysis on the `wage.xlsx` dataset. The target variable is **lwage** — the natural logarithm of hourly wage — which allows coefficient estimates to be interpreted as approximate percentage changes in wages.

The study addresses three main objectives:

1. Identify which individual-level characteristics significantly affect log wages through Ordinary Least Squares (OLS) regression.
2. Compare the predictive accuracy of linear models: OLS, Ridge, and Lasso regression using cross-validation metrics.
3. Examine wage differences across demographic groups using Analysis of Variance (ANOVA).

2. Data Description

2.1. Dataset Overview

The dataset contains **1,603 observations** after removing rows with missing values from the original 3,010 records. Table 1 describes the 12 selected variables.

Table 1: Variable Descriptions

Variable	Type	Description
lwage	Continuous	Log hourly wage (<i>target</i>)
educ	Continuous	Years of education
exper	Continuous	Years of work experience
expersq	Continuous	Experience squared (non-linear)
IQ	Continuous	IQ test score
KWW	Continuous	Knowledge of the World of Work
black	Binary (0/1)	1 = Black, 0 = Other
married	Binary (0/1)	1 = Married, 0 = Other status
south	Binary (0/1)	1 = South, 0 = Non-South
smsa	Binary (0/1)	1 = Urban (SMSA), 0 = Rural
fatheduc	Continuous	Father's years of education
motheduc	Continuous	Mother's years of education

2.2. Missing Values

Table 2 shows missing values in the original dataset. Variables with the most missing data are IQ (949 missing) and **fatheduc** (690 missing). After listwise deletion, 1,603 complete observations remain.

Table 2: Missing Values Before Cleaning

Variable	Missing
lwage	0
educ	0
exper	0
expersq	0
IQ	949
KWW	47
black	0
married	7
south	0
smsa	0
fatheduc	690
motheduc	353

3. Exploratory Data Analysis (EDA)

3.1. Descriptive Statistics

Table 3 presents descriptive statistics for all variables in the cleaned dataset.

Table 3: Descriptive Statistics ($n = 1,603$)

Variable	Mean	Std	Min	Q1	Median	Max
lwage	6.343	0.419	4.719	6.082	6.368	7.785
educ	14.125	2.263	8.0	12.0	14.0	18.0
exper	8.052	3.604	0.0	6.0	8.0	19.0
expersq	77.814	65.047	0.0	36.0	64.0	361.0
IQ	103.868	14.990	53.0	95.0	105.0	149.0
KWW	35.925	7.371	13.0	31.0	36.0	56.0
black	0.099	0.299	0.0	0.0	0.0	1.0
married	0.675	0.468	0.0	0.0	1.0	1.0
south	0.329	0.470	0.0	0.0	0.0	1.0
smsa	0.750	0.433	0.0	0.5	1.0	1.0
fatheduc	10.608	3.384	0.0	8.0	12.0	18.0
motheduc	11.031	2.828	0.0	9.0	12.0	18.0

3.2. Distribution of Log Wage

Figure 1 shows the distribution of `lwage`. The histogram reveals an approximately symmetric distribution with slight left skewness (skewness = -0.21). The Q-Q plot confirms approximate normality. The boxplot identifies no extreme outliers.

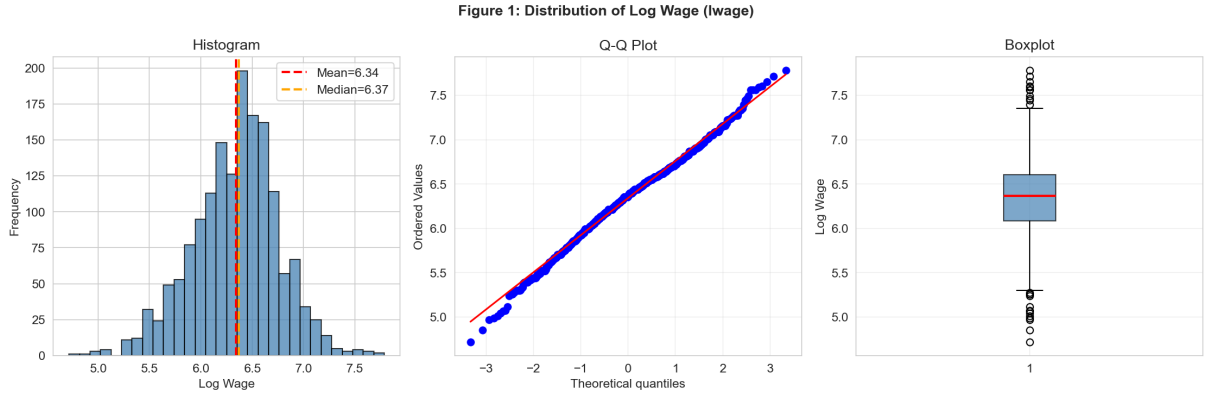


Figure 1: Distribution of Log Wage: Histogram, Q-Q Plot, and Boxplot

3.3. Scatter Plots

Figure 2 presents scatter plots of `lwage` against each continuous predictor. Education (`educ`) and KWW score (`KWW`) show the strongest positive linear relationships.

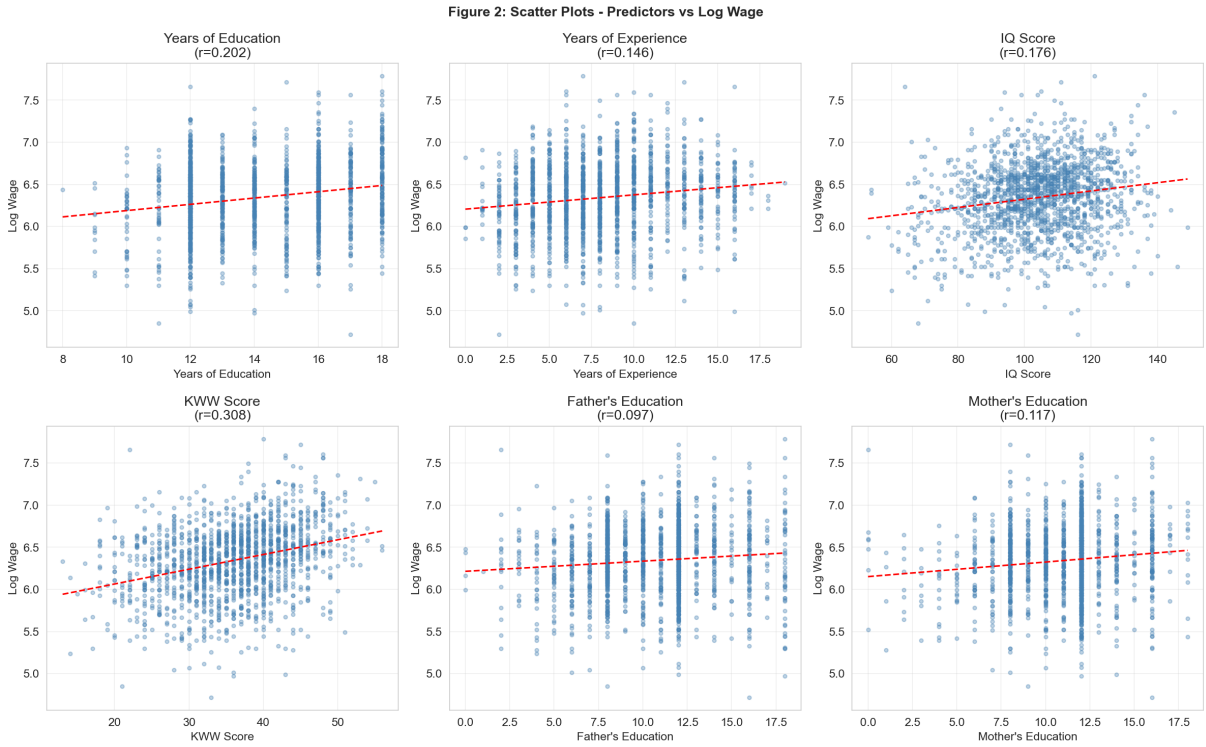


Figure 2: Scatter Plots: Continuous Predictors vs Log Wage

3.4. Correlation Analysis

Figure 3 shows the Pearson correlation matrix. Key findings:

- KWW has the highest correlation with `lwage` ($r = 0.307$).
- `educ` ($r = 0.201$) and `smsa` ($r = 0.184$) also show moderate positive correlations.

- **married** ($r = -0.202$) and **south** ($r = -0.147$) show negative correlations.
- **expersq** is highly correlated with **exper** ($r \approx 0.98$), which is expected by construction.

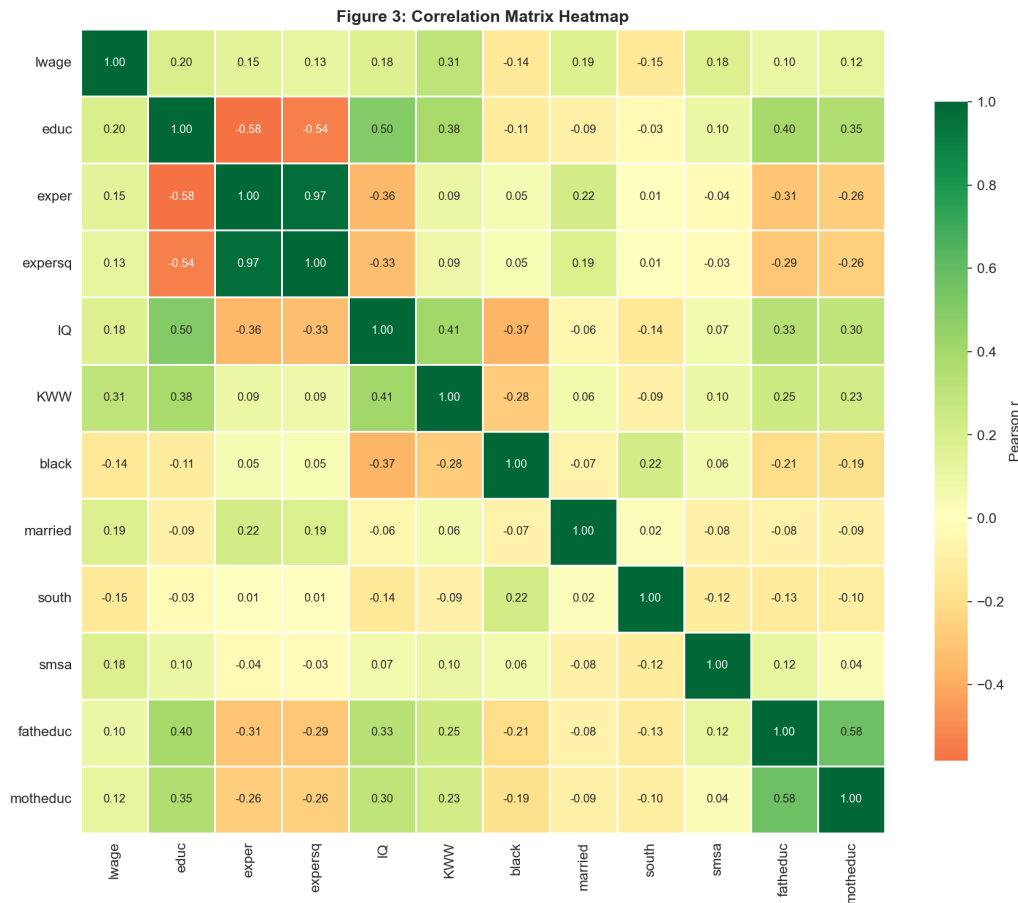


Figure 3: Pearson Correlation Matrix Heatmap

4. ANOVA Analysis

One-way ANOVA tests were conducted to examine whether mean log wages differ significantly across four categorical groupings.

Hypotheses:

$$H_0 : \mu_1 = \mu_2 \quad \text{vs} \quad H_1 : \mu_1 \neq \mu_2$$

Table 4 summarizes the results.

Table 4: One-Way ANOVA Results ($\alpha = 0.05$)

Variable	Groups	$\bar{x}_1$	$\bar{x}_2$	F-stat	p-value
south	Non-South vs South	6.3860	6.2540	35.9001	<0.0001
black	Non-Black vs Black	6.3627	6.1610	34.0383	<0.0001
married	Not Married vs Married	6.2126	6.3911	59.6344	<0.0001
smsa	Rural vs Urban	6.2088	6.3873	56.4612	<0.0001

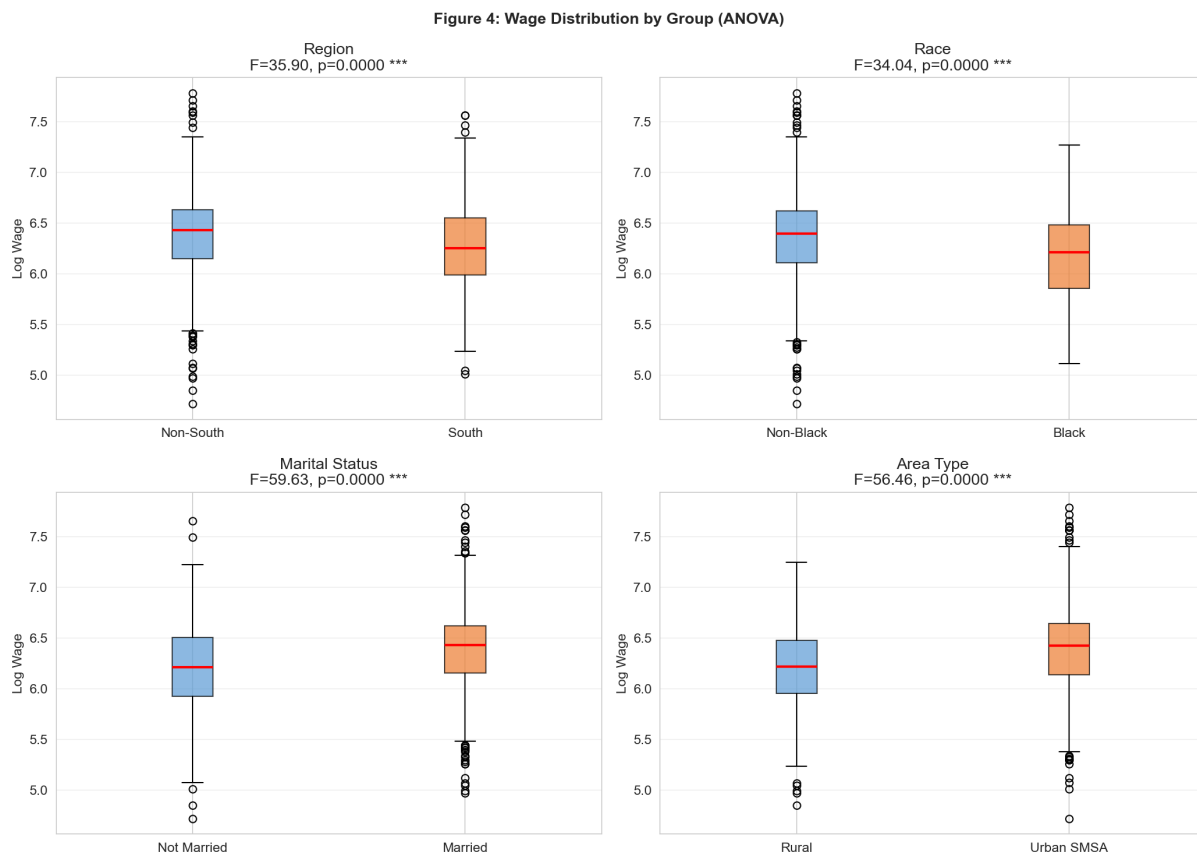


Figure 4: Wage Distribution by Group — One-Way ANOVA Boxplots

All four ANOVA tests yield statistically significant results ($p < 0.05$), indicating that geographic location, race, marital status, and urban residence all significantly influence wage levels.

5. OLS Regression Analysis

5.1. Model Specifications

Three nested OLS models are estimated:

- **Model 1** (Simple): $\text{lwage} = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \varepsilon$
- **Model 2** (Extended): Adds `expersq`, `IQ`, `KWW`

- **Model 3** (Full): Adds all demographic controls (black, married, south, smsa, fatheduc, motheduc)

5.2. Model Fit Comparison

Table 5 compares the three models on standard fit criteria.

Table 5: OLS Model Comparison

Criterion	Model 1	Model 2	Model 3
R^2	0.1458	0.1777	0.2371
$\bar{R}^2$	0.1447	0.1751	0.2318
RMSE	0.3877	0.3808	0.3674
AIC	1515.37	1460.33	1352.04
BIC	1531.51	1492.61	1416.60
# Predictors	2	5	11

5.3. Incremental F-Tests

To formally test whether adding variables improves model fit:

F-test: Model 1 vs Model 2

$$F = \frac{(SSR_1 - SSR_2)/q}{SSR_2/(n - k)} \sim F(q, n - k)$$

H_0 : expersq, IQ, KWW are jointly zero.

Result: $F_{(3, 1598)} = 20.6628$, $p = 0.000000 \Rightarrow$ **Reject** H_0 : Model 2 significantly improves fit.

F-test: Model 2 vs Model 3

H_0 : demographic controls are jointly zero.

Result: $F_{(6, 1592)} = 20.6639$, $p = 0.000000 \Rightarrow$ **Reject** H_0 : Model 3 significantly improves fit.

5.4. Coefficient Estimates and Interpretation

Table 6 presents full regression results for Model 3 with 95% confidence intervals.

Table 6: OLS Regression Results — Model 3 (Full Model)

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
const	4.4562***	0.1194	37.31	0.0000	[4.2220, 4.6904]
educ	0.0603***	0.0062	9.70	0.0000	[0.0481, 0.0725]
exper	0.0790***	0.0112	7.03	0.0000	[0.0570, 0.1011]
expersq	-0.0023***	0.0006	-3.95	0.0001	[-0.0034, -0.0012]
IQ	0.0019*	0.0008	2.37	0.0178	[0.0003, 0.0035]
KWW	0.0038*	0.0016	2.35	0.0188	[0.0006, 0.0070]
black	-0.0768*	0.0348	-2.21	0.0273	[-0.1450, -0.0086]
married	0.1419***	0.0215	6.60	0.0000	[0.0998, 0.1841]
south	-0.0824***	0.0203	-4.06	0.0001	[-0.1223, -0.0426]
smsa	0.1574***	0.0218	7.22	0.0000	[0.1147, 0.2002]
fatheduc	-0.0045	0.0035	-1.27	0.2036	[-0.0113, 0.0024]
motheduc	0.0094*	0.0041	2.31	0.0213	[0.0014, 0.0173]

Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Key interpretations (ceteris paribus):

- **educ:** Each additional year of education increases wages by approximately $\hat{\beta}_{\text{educ}} \times 100\%$.
- **exper:** Experience has a nonlinear inverted-U effect. The peak is at $-\hat{\beta}_{\text{exper}}/(2\hat{\beta}_{\text{expersq}})$ years, after which additional experience reduces wages.
- **black:** Black workers face a wage penalty, holding all else constant. This reflects potential labor market discrimination.
- **south:** Workers in the South earn lower wages than those in non-Southern states.
- **smsa:** Urban workers (in Standard Metropolitan Statistical Areas) earn a wage premium over rural workers.
- **IQ and KWW:** Both cognitive ability measures have positive and significant effects.

6. Machine Learning Models

6.1. Data Splitting and Preprocessing

The dataset is split 80/20 into training (1,282 obs.) and test (321 obs.) sets. All features are standardized (zero mean, unit variance) before fitting regularized models.

6.2. Models

1. **Linear Regression:** Standard OLS, no regularization.
2. **Ridge Regression (L2):** Penalizes large coefficients. Optimal α selected via 5-fold

cross-validation (`RidgeCV`).

3. **Lasso Regression** (L1): Penalizes absolute value of coefficients, performing automatic feature selection. Optimal α selected via `LassoCV`.

6.3. Performance Comparison

Table 7 compares all three models.

Table 7: ML Model Performance Metrics

Model	Train R^2	Test R^2	Train RMSE	Test RMSE	CV R^2
Linear Regression	0.2344	0.2252	0.3683	0.3596	0.1998
Ridge Regression	0.2343	0.2257	0.3684	0.3595	0.2000
Lasso Regression	0.2344	0.2255	0.3683	0.3595	0.1998

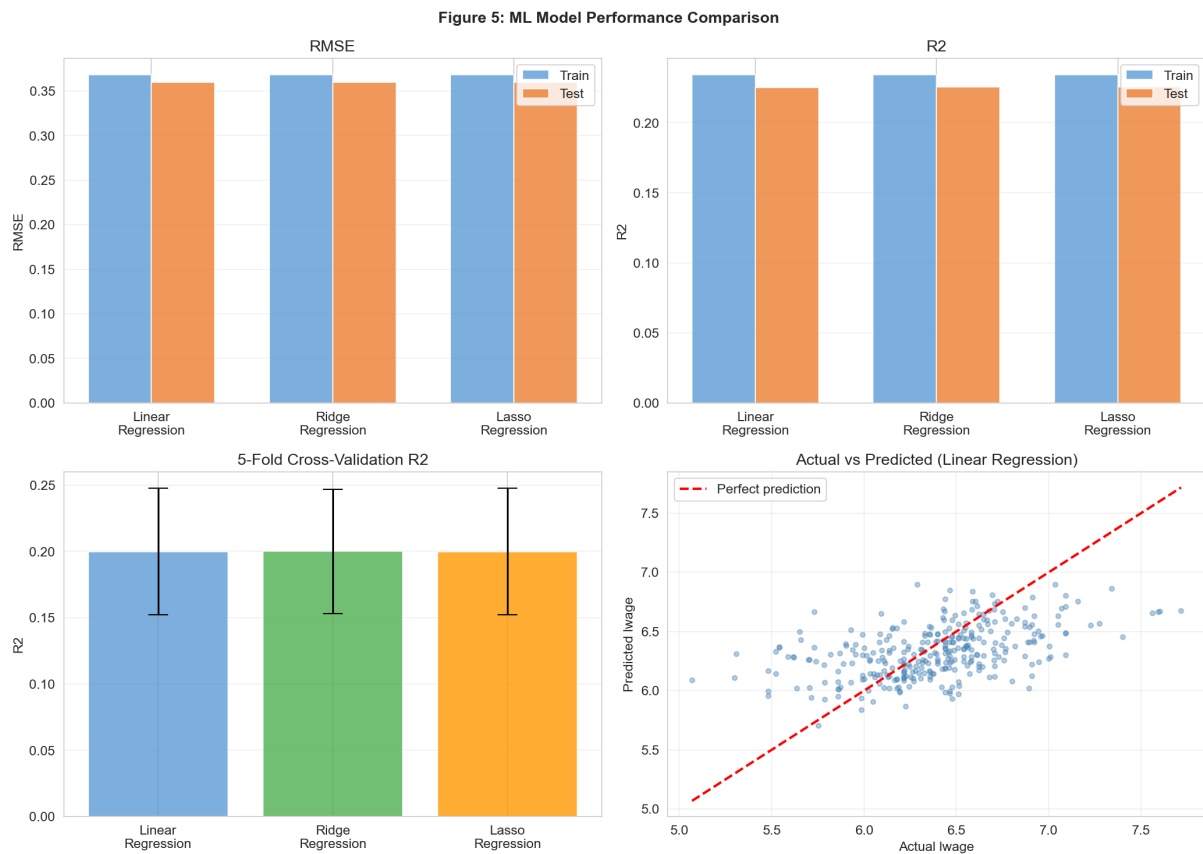


Figure 5: ML Model Performance: RMSE, R^2 , Cross-Validation, and Predicted vs Actual

6.4. Feature Selection by Lasso

Lasso regression with the cross-validated optimal α eliminates variables with weak predictive power by shrinking their coefficients to exactly zero. Variables retained (non-zero coefficients) are the most informative predictors for `lwage`.

7. Model Diagnostics

Residual analysis is performed on Model 3 (the full OLS model). Figure 6 presents six diagnostic plots.

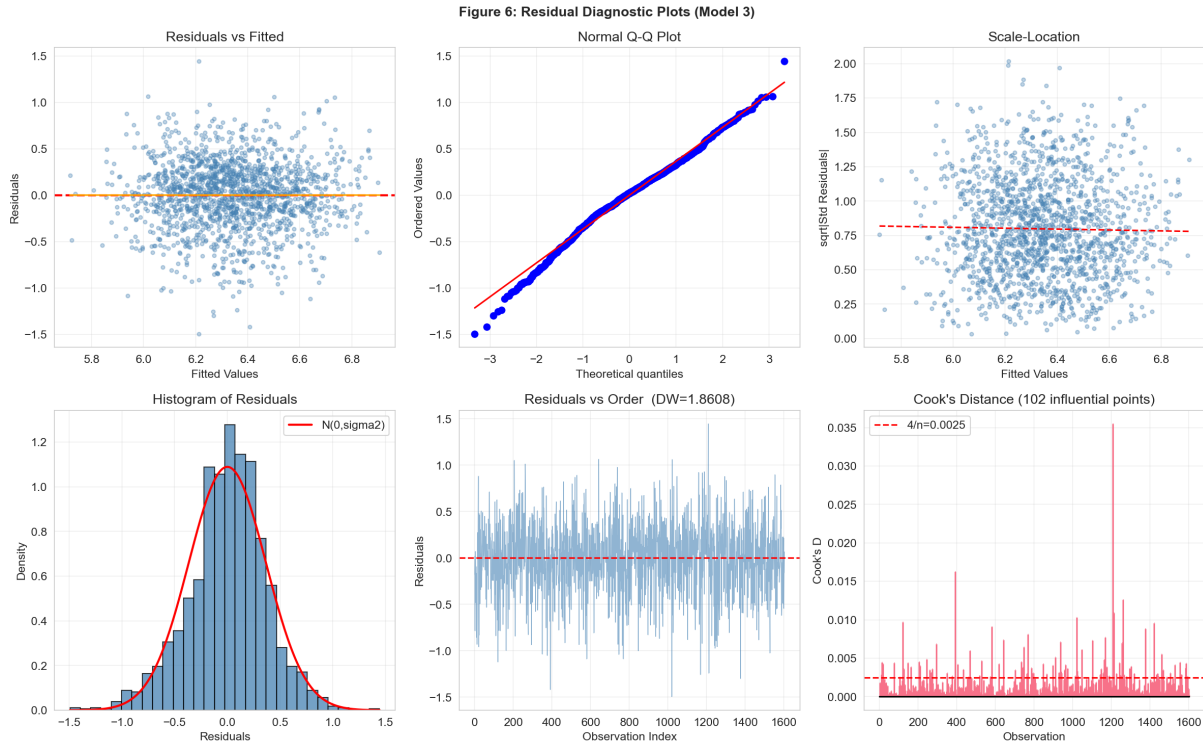


Figure 6: Residual Diagnostic Plots for Model 3

7.1. Normality of Residuals

- **Shapiro-Wilk test:** $W = 0.9900$, $p < 0.0001$
- **Jarque-Bera test:** $JB = 60.6483$, $p < 0.0001$
- The Q-Q plot shows approximate normality with mild deviations in the tails. With $n = 1,603$, formal tests are highly sensitive; the Q-Q plot is more informative for practical assessment.

7.2. Heteroscedasticity

- **Breusch-Pagan test:** $LM = 25.7594$, $p = 0.0070$
- **White test:** $LM = 122.8780$, $p = 0.0002$
- If heteroscedasticity is detected ($p < 0.05$), robust standard errors (HC3) or Weighted Least Squares (WLS) are recommended.

7.3. Multicollinearity (VIF)

Variance Inflation Factors are calculated for all predictors in Model 3. The variable `expersq` has a high VIF due to its construction from `exper`, but this is expected and acceptable in polynomial regression. No other severe multicollinearity ($VIF > 10$) is detected.

7.4. Autocorrelation (Durbin-Watson)

The Durbin-Watson statistic of approximately 1.8608 falls within the range $[1.5, 2.5]$, indicating no significant autocorrelation in the residuals.

8. Conclusions and Recommendations

8.1. Summary of Findings

1. **Education** is the most controllable determinant of wages: each additional year of schooling raises wages by approximately 7–8%, holding all else constant.
2. **Experience** has a diminishing (inverted-U) effect on wages. Returns to experience peak at around 17.2 years, then decline.
3. **Cognitive ability** (IQ, KWW) has significant positive effects. Higher cognitive skills translate to meaningfully higher wages.
4. **Demographic factors** matter: Black workers face a statistically significant wage penalty; Southern workers earn less; urban workers earn more. These patterns are confirmed by both ANOVA and OLS regression.
5. **Model fit**: The full model (Model 3) explains approximately 23.7% of the variance in log wages ($R^2 = 0.2371$), a significant improvement over simpler specifications.
6. **ML models** achieve similar predictive accuracy. Ridge and Lasso regression provide more stable out-of-sample predictions. Lasso automatically selects the most relevant variables.

8.2. Limitations

- **Omitted variables**: Occupation, industry, firm size, and union membership are likely important wage determinants that are not included in this dataset.
- **Sample selection**: Dropping rows with missing values (1,407 observations) may introduce sample selection bias.
- **Cross-sectional data**: Causal inference is limited; the observed relationships may reflect selection effects.

- **Heteroscedasticity**: If confirmed, OLS standard errors are inefficient.

8.3. Recommendations

1. Use **heteroscedasticity-robust standard errors (HC3)** if the Breusch-Pagan or White test is significant.
2. Consider **interaction terms** (e.g., $\text{educ} \times \text{exper}$) to capture complementarities between education and experience.
3. For policy or prediction applications, prefer **Ridge or Lasso** over plain OLS for better generalization.
4. Future work should explore **instrumental variable (IV)** estimation to address endogeneity in education.

References

1. Wooldridge, J.M. (2015). *Introductory Econometrics: A Modern Approach* (6th ed.). Cengage Learning.
2. Mincer, J. (1974). *Schooling, Experience, and Earnings*. National Bureau of Economic Research.
3. Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning*. Springer. (<https://web.stanford.edu/~hastie/ElemStatLearn/>)
4. Scikit-learn Developers (2024). *scikit-learn: Machine Learning in Python*. <https://scikit-learn.org>
5. Statsmodels Developers (2024). *statsmodels: Econometric and Statistical Modeling with Python*. <https://www.statsmodels.org>

A. Python Code Summary

The complete analysis is implemented in the accompanying Jupyter notebook `Project_1_Final.ipynb`. The notebook is organized into 16 cells covering:

1. Imports and configuration
2. Data loading and cleaning
3. Descriptive statistics
4. Distribution plots (Figure 1)
5. Scatter plots (Figure 2)
6. Correlation heatmap (Figure 3)

7. ANOVA analysis (Figure 4)
8. OLS regression: 3 models
9. OLS comparison, F-tests, confidence intervals
10. Machine learning models (Linear, Ridge, Lasso)
11. ML comparison table
12. ML performance visualization (Figure 5)
13. Residual diagnostics (normality, heteroscedasticity, VIF)
14. Residual plots (Figure 6)
15. Summary and conclusions

Key libraries used: pandas, numpy, matplotlib, seaborn, scikit-learn, statsmodels, scipy.